

RoG-SAM: A Language-Driven Framework for Instance-Level Robotic Grasping Detection

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Abstract—Robotic grasping is a crucial topic in robotics and computer vision, with broad applications in industrial production and intelligent manufacturing. Although some methods have begun addressing instance-level grasping, most remain limited to predefined instances and categories, lacking flexibility for open-vocabulary grasp prediction based on user-specified instructions. To address this, we propose RoG-SAM, a language-driven, instance-level grasp detection framework built on Segment Anything Model (SAM). RoG-SAM utilizes open-vocabulary prompts for object localization and grasp pose prediction, adapting SAM through transfer learning with encoder adapters and multi-head decoders to extend its segmentation capabilities to grasp pose estimation. Experimental results show that RoG-SAM achieves competitive performance on single-object datasets (Cornell and Jacquard) and cluttered datasets (GraspNet-1Billion and OCID), with instance-level accuracies of 91.2% and 90.1%, respectively, while using only 28.3% of SAM's trainable parameters. The effectiveness of RoG-SAM was also validated in real-world environments.

Index Terms—Robotic vision, grasp detection, language-guided detection, segment anything model, fine-tuning.

I. INTRODUCTION

ROBOTIC grasping is a key focus in robotics and computer vision, with extensive applications in industrial production and intelligent manufacturing [1]. Traditional methods predominantly predict feasible grasp poses for objects in a scene without explicitly identifying individual instances or categories [2], [3], [4], [5]. Although some approaches have begun addressing instance-level grasping [6], [7], they are often

constrained by predefined instances or categories, lacking the flexibility for diverse, open-vocabulary scenarios. Such limitation reduces their effectiveness in tasks requiring precise interactions with specific targets, particularly in complex, multi-object environments [8], [9].

Real-world scenarios present significant challenges, including grasping unseen objects [10] and managing cluttered environments [6]. While humans can intuitively interpret object characteristics and make adaptive decisions, robots with limited prior knowledge struggle in these complex settings. Developing a universal grasp detection framework that enables robots to perform targeted grasps in novel environments guided by user input remains a demanding task, requiring comprehensive visual understanding of real-world contexts. Instance-aware grasp prediction enhances real-time, interactive robotic operations by offering greater flexibility, which are particularly valuable in applications like assembly line sorting and warehouse automation [11].

Existing grasping methods often rely on CNN architectures and their specialized variants for grasp pose prediction [12], [13], [14]. These approaches perform pixel-level predictions by regressing confidence scores for each image position. However, their reliance on domain-specific datasets limits their generalization to diverse visual patterns, restricting their applicability beyond predefined scenarios. The emergence of foundational vision models has introduced new possibilities for visual robotics. Among these, the segment anything model (SAM) [15] has gained significant attention due to its zero-shot and few-shot segmentation capabilities. SAM predicts multiple segmentation masks based on user-provided prompts, which offers a promising foundation for instance-level interactive grasp prediction in robotics. However, despite its exceptional performance on natural images, recent studies have shown that SAM can underperform in certain downstream tasks due to a lack of specific knowledge [16], [17]. Adapting SAM for grasp detection poses two major challenges: addressing domain-specific disparities and transitioning from semantic segmentation toward predicting interactive object components, such as grasp angles and widths, critical for robotic grasping. As emphasized by [18], the effectiveness of grasping models heavily depends on training data, highlighting the importance of data domain transfer in adapting large models for downstream applications.

In this paper, we introduce RoG-SAM, an instance-level object grasp detection algorithm based on SAM. RoG-SAM utilizes open-vocabulary user prompts to predict grasp poses for

Received 24 August 2024; revised 22 November 2024 and 9 January 2025; accepted 10 January 2025. Date of publication 3 April 2025; date of current version 28 May 2025. This work was supported in part by the National Key R&D Program of China under Grant 2021YFB1714800 and in part by the National Natural Science Foundation of China under Grant 62173034, Grant U23B2059, and Grant 62088101. The guest editor coordinating the review of this article and approving it for publication was Prof. Mengshi Qi. (Corresponding author: Gang Wang.)

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A demonstration video is available at <https://www.youtube.com/playlist?list=PL7et4nGJAImLGYtsJbglGbX1lhacA2dy..>

Digital Object Identifier 10.1109/TMM.2025.3557685

specific objects. Our model first uses these textual prompts to identify object locations (bounding boxes), which are subsequently input into a modified SAM model for grasp pose prediction. We adapt the pretrained SAM model for grasping tasks via transfer learning. By incorporating adapter and multi-head decoder modules, we extend SAM's capabilities from semantic segmentation to the prediction of grasp poses. The performance of RoG-SAM is evaluated on both single-object-centric datasets [3], [19] and multi-object datasets [6], [7] to assess its grasp pose regression and instance-level recognition capabilities. Additionally, we test the model in real-world scenarios, focusing on object-specific grasping tasks for unseen objects and cluttered environments. Experimental results show that RoG-SAM delivers competitive performance across four public datasets, demonstrating its effectiveness in interactive instance-level object grasping while utilizing only 28.3% of the trainable parameters of the full SAM model.

In summary, the contributions of this paper are as follows:

- 1) We propose RoG-SAM, a language-driven framework for interactive, instance-level robotic grasping. By integrating encoder adapters and multi-head decoder modules, RoG-SAM extends the Segment Anything Model (SAM) to grasp pose prediction, with only 28.3% of SAM's original trainable parameters.
- 2) Extensive experiments demonstrate RoG-SAM's robust generalization across diverse settings, achieving competitive accuracies of 99.4%, 94.6%, 91.2%, and 90.1% on the Cornell, Jacquard, GraspNet-1Billion, and OCID datasets, respectively. Real-world tests on a UR5 robot further confirm its effectiveness in grasping unseen objects, with accuracies of 97.5% in single-object and 94.6% in cluttered environments.

II. RELATED WORK

A. Grasp Detection

Robotic grasp detection has been a key area in robotics research, largely focused on predicting grasp poses using RGB or RGB-D inputs. Early methods employed CNNs for end-to-end, pixel-level grasp pose prediction [13], [14].

Recent advancements have integrated sub-visual tasks into their pipeline to achieve instance-level detection, including object segmentation [6], local classification [20], and region of interest (ROI) analysis [21]. These methods generally adopt a two-stage pipeline. The initial stage predicts potential grasp poses for all objects within the scene. In the second stage, the detection is refined to locate the target object. Typically, target locations are represented with dense representations, including semantic masks or localized image patches. The final grasp pose is selected based on these detection results, as depicted in Fig. 1(a).

While these methods are effective in various scenarios, they generally predict grasp poses for all objects in an image, requiring post-processing to identify target grasp. This process relies on location detectors, which are often limited to predefined object classes. Such constraints pose challenges when dealing with unseen or similar objects, especially in cluttered environments where detection models may fail, leading to inaccurate grasp predictions when distinguishing between instances with similar

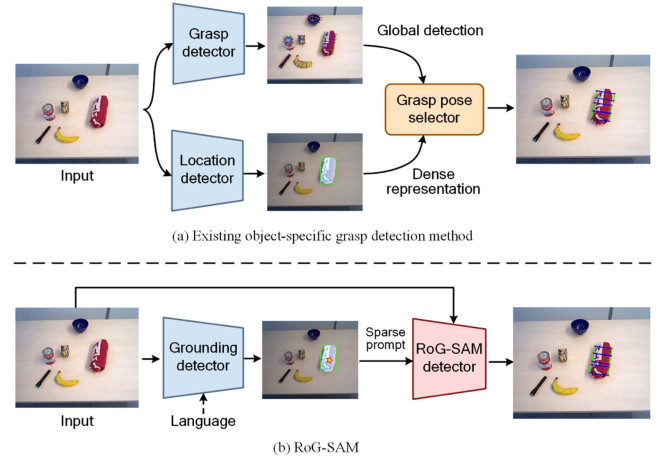


Fig. 1. Comparison between existing object-specific grasp detection methods and the RoG-SAM framework when detecting the grasp poses of a towel. Unlike existing two-stage methods that rely on a global grasp detector and post-processing modules to select grasp poses, RoG-SAM employs a single-stage pipeline that directly predicts target grasp poses using sparse prompts, such as bounding boxes and points generated by a grounding detector.

class. This challenge is also evident in language-driven robotic navigation tasks [22], [23].

To address these challenges, we developed a grasp prediction framework that directly predicts grasp poses for specific instances from a single image, as shown in Fig. 1(b). The framework uses sparse object location representations in the form of bounding boxes and coordinates as prompts to guide the prediction process. This instance-focused approach provides direct grasping suggestions for specific targets.

B. Vision Foundation Models

Vision foundation models have emerged as a key innovation in computer vision, similar to the transformative role of foundational models in natural language processing [24]. Pre-trained on large-scale image datasets using supervised or semi-supervised learning, these models excel in generalizing across diverse visual tasks, making them highly valuable for applications like robotic grasping. A prime example is CLIP [25], which leverages open-vocabulary text as a supervisory signal to align image-text embeddings within a shared multimodal space. This alignment enables CLIP to achieve strong performance in zero-shot learning, particularly in recognizing unseen categories. CLIP's functionality spans a range of tasks, such as image classification [26], and it serves as a foundational model for systems like DALL-E [27], which extends CLIP's framework to generate images from text descriptions through VQ-VAE [28]. Other vision foundation models like DINO [29], use vision transformers to extract visual features from unlabeled data. The self-emerging attention patterns in DINO have shown significant potential and applicability across multiple visual tasks. Grounding DINO [30] extends DINO by incorporating Grounded Pre-Training [31] techniques to handle open-set object detection tasks. In practical applications, Grounding DINO can discover and identify previously unseen object categories based on textual descriptions or visual cues. Similarly, SAM [15], through semi-supervised training, extracts generalizable segmentation features from

extensive image and segmentation data. Its Transformer-based architecture efficiently executes image segmentation tasks based on various prompts. In robotic tasks, where accurate recognition and localization of target objects in complex and diverse environments are crucial, leveraging vision foundation models can significantly enhance object detection, recognition, and segmentation, thereby increasing the success rate and flexibility of robotic grasping operations.

C. Robotic Grasping With Vision Foundation Models

Recent efforts have utilized vision foundation models to enhance the perception and interaction capabilities of robotic grasping systems. For instance, [32] employs SAM as a zero-shot segmenter, utilizing a set of uniformly distributed point prompts to segment masks for all objects in an image. The method then queries the segmented masks against a template library to infer its grasp pose. By leveraging SAM's semantic awareness and the PMF classifier [33], this approach can grasp specific objects based on fixed prompts, demonstrating zero-shot and few-shot capabilities. However, fixed, uniformly distributed point prompts may not include all objects in the scene, and example images of target objects are not always readily available. Another approach, [34], introduces a CLIP-based segmentation model that learns grasping strategies based on user prompts by exploiting CLIP's image-language alignment capability. This framework uses supervised data in the form of $(\mathbf{I}, \mathbf{L}) \rightarrow \mathbf{G}$, where $\mathbf{I}$ represents the image, $\mathbf{L}$ denotes the language prompt, and $\mathbf{G}$ signifies the corresponding grasp pose, for end-to-end training. Due to the scarcity of grasp pose datasets with language descriptions, this work re-annotated the OCID dataset. However, the high cost associated with re-annotating large datasets is often prohibitive.

In this work, we present a novel language-driven grasping framework that leverages a grounding detector to generate bounding box and point prompts from text inputs, enabling instance-level training on existing grasping datasets without the need for additional manual annotations.

III. METHODOLOGY

This section introduces the implementation of the proposed open-vocabulary grasping framework, RoG-SAM. Subsection A defines the grasping problem, Subsection B outlines the RoG-SAM architecture, including its two-stage pipeline and lightweight strategies, and Subsection C describes the loss function adopted in this work.

A. Problem Statement

In robotic grasping, the core challenge is determining the optimal position, orientation, and aperture of the gripper to achieve a grasp on target object. Typically, a grasp pose is represented by a two-finger parallel gripper with a 4-degree-of-freedom (4-DoF) vector, which balances simplicity and functionality effectively [35].

Given an RGB image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ and a language prompt $\mathbf{T}$ (e.g., "fruit" or "cracker box"), the model aims to predict

feasible grasp poses for the object specified by the prompt:

$$\mathbf{G} = F(\mathbf{I}, \mathbf{T}; \phi) \quad (1)$$

where F represents the RoG-SAM model, and ϕ denotes its trainable parameters. The output $\mathbf{G} = \{G_i\}_{i=1}^N$ denotes all potential grasp poses for the single object specified by the prompt, where N is the number of grasp poses predicted for the object. Each grasp pose $G_i = (x_i, y_i, \theta_i, w_i)$ defines the i -th grasp for the object. In this formulation, (x_i, y_i) specifies the 2D location of the grasp, θ_i defines the gripper's orientation, and w_i denotes the required gripper aperture to grasp the object.

B. The RoG-SAM Framework

The RoG-SAM framework consists of four core components: the grounding detector, the image encoder, the prompt encoder, and the mask decoder, as illustrated in Fig. 2(a). These components work together within a two-stage approach, which integrates a **grounding stage** and a **grasp prediction stage** to achieve instance-level robotic grasping. Additionally, we employ **lightweight strategies** to optimize computational efficiency. The following subsections detail the grounding and grasp prediction stages as well as the lightweight strategies applied.

Grounding stage: In the grounding stage, sparse position prompts are generated to guide the prompt encoder in identifying and locating a target object within an image $\mathbf{I}$ based on a given text prompt $\mathbf{T}$. These sparse prompts are represented as a single 2D bounding box $\mathbf{b}$ and points sampled from a semantic mask $\mathbf{m}$ for the target object. The grounding process can be formalized as $(\mathbf{b}, \mathbf{m}) = g(\mathbf{I}, \mathbf{T}; \varphi_g)$, where g denotes the grounding detector model and φ_g represents its parameters. In our implementation, we utilize GroundedSAM [36] as the grounding detector, configured with ViT-B variant of SAM and Swin-T variant of GroundingDINO, to generate bounding box and point prompts from the input text.

Grasp prediction stage: Following the grounding stage, the grasp prediction stage employs an enhanced SAM model to predict grasp poses, represented as $\mathbf{G} = S(\mathbf{I}, \mathbf{b}, \mathbf{p}; \varphi_S)$, where φ_S denotes the model parameters, $\mathbf{b}$ represents the bounding box prompt for the target object, and $\mathbf{p} = \{p_k\}_{k=1}^K$ represents the set of point prompts, with each point prompt p_k sampled from the semantic mask $\mathbf{m}$. Here, K is the number of sampled points.

The enhanced SAM model $S(\cdot)$ first predicts three distinct masks for the target object: position mask, width mask, and angle mask. Grasp poses are generated by identifying the maximum position of position mask, producing one or more grasp candidates. The roles of these three pose-representing masks are as follows:

- *Position mask:* A single-channel affordance heatmap that indicates grasp confidence at each pixel, used to determine the grasp center.
- *Width mask:* A single-channel heatmap that represents gripper width features at each pixel. Width values are initially expressed as pixel lengths and are normalized to $[0, 1]$ to ensure training stability.
- *Angle mask:* A multi-channel approach ($C = 180$) is employed to represent angular ranges. These masks depict the

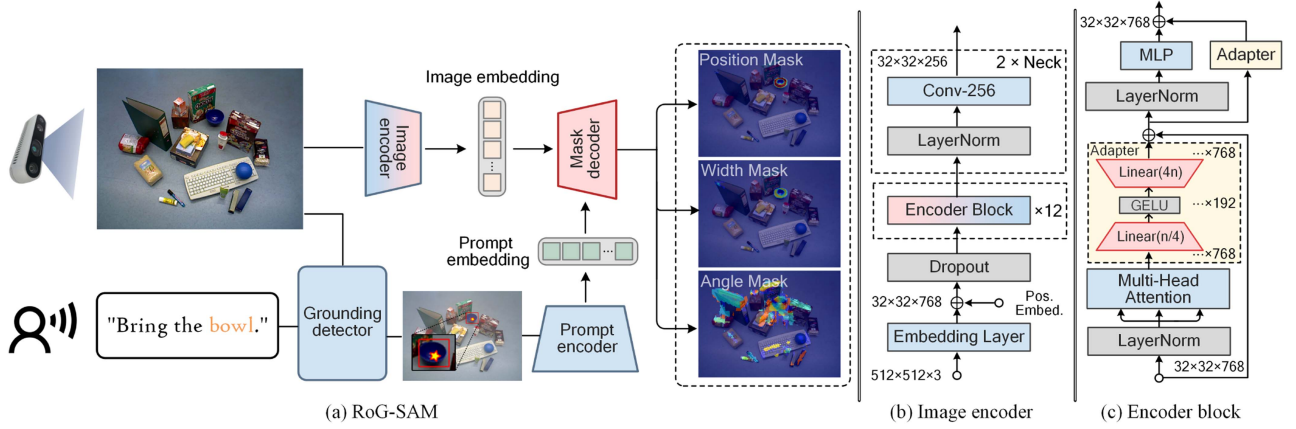


Fig. 2. Overview of RoG-SAM framework and the image encoder. (a) Architecture of the RoG-SAM framework. For visualization purposes, the multi-channel angle mask produced by RoG-SAM is merged into a single-channel mask, while the position and width masks are overlaid directly onto the RGB image. (b) Structure of the image encoder, which consists of a transformer encoder with 12 encoder blocks and a Neck module stacked twice. (c) Configuration of an individual encoder Block, with the dashed area indicating the adapter module. In the entire figure, frozen parameters or modules are shown in blue, functional layers (dropout, layernorm) are depicted in gray, and trainable layers or modules are highlighted in warm or gradient warm colors.

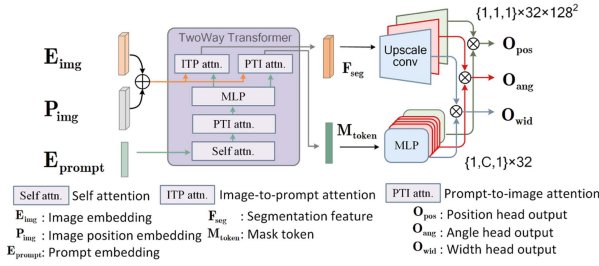


Fig. 3. Structure of the mask decoder in RoG-SAM.

grasp angle at each pixel, where each channel corresponds to an angular range of $\frac{2\pi}{C}$. The class for a given angle θ at a pixel is determined by $\lceil \frac{\theta C}{2\pi} \rceil$, with the confidence at that pixel set to 1.

The mask decoder of RoG-SAM is designed with multiple segmentation heads to predict various features of a grasp pose, as illustrated in Fig. 3. The decoder processes image embeddings E_{img} generated by the image encoder, along with corresponding position embeddings P_{img} and prompt embeddings E_{prompt} from the prompt encoder. These embeddings are refined through a two-way transformer, which employs mutual attention modules between the image and prompt embeddings, producing segmentation features F_{seg} and mask tokens M_{token} . These outputs are subsequently fed into upscaling convolution modules and MLPs.

During the prediction phase, features from the upscaling convolution module are combined with mask tokens, which are processed by a set of MLPs through dot product operations. This results in a final set of $C + 2$ masks. The final predictions are mathematically formulated as follows:

$$O_{pos} = U_{pos}(F_{seg}) \cdot M_{pos}(M_{token}) \quad (2)$$

$$O_{ang} = \sum_{i=1}^C (U_{ang}(F_{seg}) \cdot M_{ang}^i(M_{token})) \quad (3)$$

$$O_{wid} = U_{wid}(F_{seg}) \cdot M_{wid}(M_{token}) \quad (4)$$

where O_{pos} , O_{ang} , and O_{wid} represent the output masks for position, angle, and width predictions, respectively. U represents the upscaling convolution modules for each prediction head, while M corresponds to the MLPs that process the mask tokens. Specifically, the angle prediction O_{ang} is derived by summing the contributions from C distinct MLPs, denoted as M_{ang}^i , where each M_{ang}^i corresponds to a specific angular range.

Lightweight strategies: SAM has acquired extensive visual knowledge through training on the SA-1B dataset [15], which encompasses 11 million images and 1 billion masks. To adapt SAM for robotic grasping while minimizing training costs, we developed RoG-SAM based on the SAM-ViT-base model.

For efficient transfer learning, we froze the weights of the grounding detector and the prompt encoder, as they must remain versatile to accommodate diverse image and language inputs, as indicated by the blue color in Fig. 2(a). The mask decoder, on the other hand, was fully trained due to structural modifications, as highlighted in warm colors. The image encoder, which contains the majority of SAM's parameters and is illustrated in Fig. 2(b), was fine-tuned using the parameter-efficient fine-tuning (PEFT) method [16]. Specifically, we incorporated trainable adapter modules into each encoder block within image encoder. These adapters were placed after multi-head attention blocks and within the residual paths of the MLPs, as shown in Fig. 2(c) and marked in warm colors. The adapter modules compress and expand feature vectors by a factor of four, utilizing the GELU activation function [37] for feature mapping. This approach allows the transformer blocks to adjust their parameters specifically for the grasping task.

C. Loss Function

Robotic grasping datasets typically have sparse pixel-level annotations, leading to a class imbalance where most pixels represent non-graspable regions and only a few denote graspable points. This imbalance is more pronounced in multi-channel angle mask predictions, where each channel corresponds to a specific angular range. To address this, we use a weighted BCE

loss to calculate segmentation errors for each prediction head

$$\mathcal{L} = -\frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \left[h_{ij} w \log(\sigma(\hat{h}_{ij})) + (1 - h_{ij}) \log(1 - \sigma(\hat{h}_{ij})) \right] \quad (5)$$

where H and W represent the height and width of prediction masks, which are set to 512 in our implementation. The variables h_{ij} and $\hat{h}_{ij}$ denote the ground truth and the predicted value at position (i, j) within each mask, respectively. The Sigmoid activation function is represented by $\sigma(\cdot)$. The weight assigned to positive samples, w , is calculated as follows

$$w = \frac{HW - \sum_{i=1}^H \sum_{j=1}^W h_{ij}}{\sum_{i=1}^H \sum_{j=1}^W h_{ij}}. \quad (6)$$

This weighting ensures the loss function remains sensitive to the sparsely annotated graspable regions, which is crucial for the training process in our task.

IV. EXPERIMENT

This section presents an overview of the training configuration and experimental setup for RoG-SAM. Subsection A specifies the hyperparameters and hardware configurations employed during training. Subsection B introduces the benchmark grasping datasets used to assess the performance of our method, alongside the evaluation metrics. Subsection C discusses the experimental results obtained on each benchmark dataset, including findings from the ablation studies conducted.

A. Training Configuration

RoG-SAM was fine-tuned from the pretrained SAM weights. We used the Adam optimizer with a batch size of 4, starting with an initial learning rate of 3×10^{-4} , which was reduced by a factor of 0.6 every 10 epochs. Training was performed on an Nvidia GeForce RTX 4090 GPU. The input images were preserved at their original resolution for each dataset, without any cropping-like preprocessing applied. If necessary, images were padded to a square format, and the output masks were generated at a resolution of 256. To improve model robustness, data augmentation techniques such as random cropping, rotation, and scaling were employed during training.

B. Grasp Dataset

We selected four widely used benchmark datasets to ensure a comprehensive evaluation of our model: Cornell [19], Jacquard [3], GraspNet-1Billion [7], and OCID [6]. The Cornell and Jacquard datasets are object-centered, with each scene containing a single object centered in the image, which facilitates evaluating grasp detection accuracy in isolated settings. In contrast, the GraspNet-1Billion and OCID datasets feature cluttered scenes with multiple objects, providing a more challenging environment to assess the model's ability for instance-aware grasp prediction. A brief overview of these datasets follows.

Cornell Dataset: The Cornell grasp dataset is one of the most extensively utilized and foundational datasets for the evaluation of grasping algorithms. It comprises 1,035 images at a resolution of 640×480 pixels, depicting 240 distinct household objects. The dataset includes ground truth annotations as rectangles representing grasp poses for each object, totaling 8,019 grasp poses. The Cornell dataset supports two primary splits: image-wise and object-wise. The image-wise split allocates different poses of same object between training and testing sets, which primarily tests the model's ability to generalize across varying poses of known objects. The object-wise split divides different object instances between training and testing, thereby evaluating the model's capacity to generalize to unseen objects.

Jacquard Dataset: The Jacquard dataset is a large-scale grasp dataset, derived from a subset of objects within the ShapeNet repository [38]. The dataset encompasses 54,000 images and approximately 1.1 million annotated grasp poses. We randomly selected 80% of the samples for training and the remaining 20% for testing model performance.

GraspNet-1Billion: The GraspNet-1Billion dataset is a comprehensive benchmark designed for general object grasping, comprising 190 cluttered scenes captured by RGB-D cameras. It includes 97,280 images, each annotated with accurate 6D object poses and 4-DoF grasp poses represented as rectangles. The dataset spans 88 distinct objects and over 1.1 billion annotated grasp poses, making it the most extensive resource for evaluating grasping models in realistic and challenging environments.

OCID Grasp Dataset: The OCID Grasp dataset extends the OCID object segmentation dataset [39], making it suitable for robotic grasping tasks by incorporating ground truth candidate grasp poses, represented as rectangles. In addition, OCID provides detailed semantic masks and object class labels, offering comprehensive instance-level context. The dataset is particularly focused on cluttered indoor environments, containing 11,000 images and approximately 75,000 annotated grasp poses. We utilized the official training and testing splits.

We evaluate the accuracy of our grasp detection algorithm using two established metrics: Jaccard index and angle prediction error. A predicted grasp pose is considered accurate if it satisfies the following conditions.

- c1) The angular difference between the predicted grasp angle and the ground truth angle is less than 30° .
- c2) The Jaccard index between the predicted grasp rectangle and the ground truth grasp rectangle exceeds 0.25.

The Jaccard index is a measure of similarity between the predicted grasp rectangle and the ground truth, which is defined as

$$J(\bar{G}, G_{gt}) = \frac{|\bar{G} \cap G_{gt}|}{|\bar{G} \cup G_{gt}|} \quad (7)$$

where $\bar{G}$ denotes the predicted pose and G_{gt} represents the ground truth grasp pose.

C. Experimental Results

Unless otherwise noted, the results of comparative methods were taken from their publications. The fine-tuning epochs

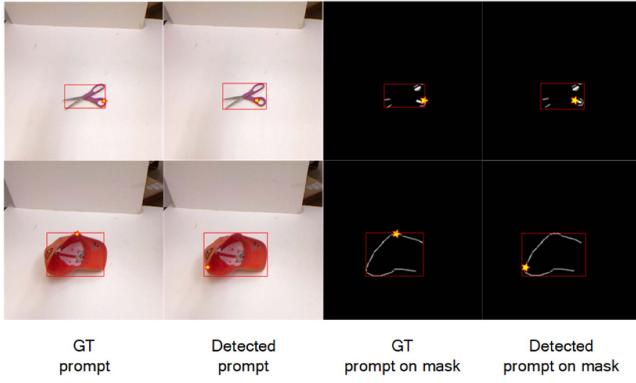


Fig. 4. Comparison between GT prompts and prompts detected by grounding detector.

TABLE I
COMPARISON OF RoG-SAM WITH RECENT WORKS ON CORNELL DATASET

Method	Input	IW(%)	OW(%)	Year	IA
Chalvatzaki et al. [40]	D	91.1	—	2020	✗
GR-ConvNet [2]	RGB	96.6	95.5	2021	✗
GR-ConvNet [2]	RGBD	97.7	96.6	2021	✗
Cheng et al. [41]	RGB	96.2	—	2022	✗
SKGNet [4]	RGBD	99.1	98.4	2023	✗
BA-Grasp [5]	RGBD	97.8	96.7	2024	✗
RoG-SAM-GT	RGB	99.4	98.8	—	✓
RoG-SAM-Det	RGB	99.4	98.5	—	✓

for RoG-SAM were capped at 30, 20, 20 and 50 for Cornell, Jacquard, GraspNet-1Billion and OCID datasets respectively to mitigate overfitting, with the best results reported within these constraints. We introduced the GT prompt method to enhance training efficiency by directly generating bounding boxes and point prompts as enclosing rectangles around grasp locations or semantic masks (if provided), with points randomly sampled from these annotations.

Fig. 4 compares the two prompt generation methods using scissors and a cap as examples. The first column shows GT prompt, and the second displays detected prompts. The third and fourth columns overlay grasp position annotations on these prompts. Testing on Cornell dataset revealed comparable training performance for both methods. Thus, the GT prompt was adopted for large-scale datasets to accelerate training.

We compared RoG-SAM with recent works on Cornell dataset, with the results presented in Table I, with GT prompt represented as RoG-SAM-GT and detected prompt represented as RoG-SAM-Det. IA refers to Instance Awareness, indicating whether the method operates at instance level. Using GT prompt, RoG-SAM achieved accuracy of 99.4% and 98.8% for image-wise and object-wise dataset splits, respectively, demonstrating strong competitiveness against current RGB and RGB-D based algorithms. As shown in rows 7 and 8 of Table I, the detected prompts resulted in only a 0.3% decrease in grasp accuracy compared to GT prompt in object-wise split, indicating two methods yield almost equivalent performance.

Fig. 5 showcases detections using GT prompts for various objects in Cornell dataset. The first row shows input RGB images

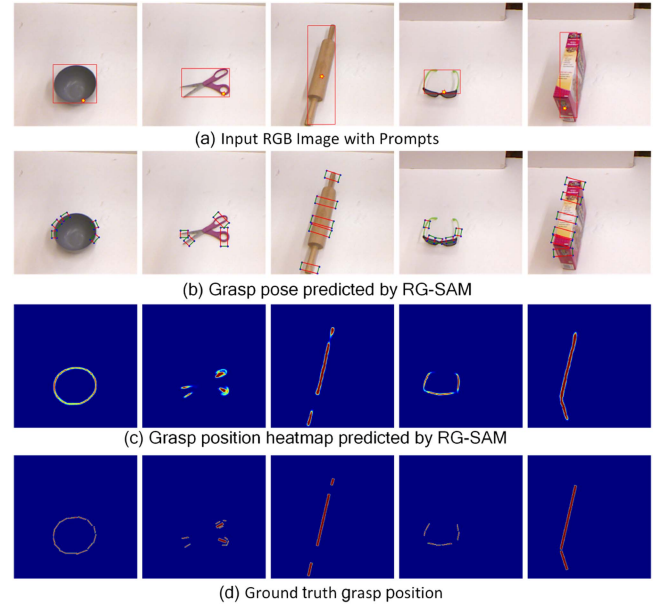


Fig. 5. Detection results for representative samples from Cornell dataset.

TABLE II
COMPARISON OF RoG-SAM WITH RECENT WORKS UNDER DIFFERENT JACCARD THRESHOLD ON JACQUARD DATASET

Method	Input	Jaccard threshold				IA
		0.20	0.25	0.30	0.35	
Song et al. [42]	RGB	—	91.5	89.7	87.3	✗
Depierre et al. [43]	RGB	88.3	85.7	82.6	78.7	✗
GR-ConvNet [2]	RGBD	—	94.6	—	—	✗
Cao et al. [44]	D	—	95.6	—	—	✗
Grasp-Det [6]	RGB	—	92.9	91.3	89.0	✓
SSG [45]	RGBD	—	91.8	90.0	88.5	✓
NCIGrasp [46]	RGBD	—	96.1	—	—	✗
RoG-SAM-GT	RGB	94.8	94.6	94.0	93.6	✓

with their corresponding box and point prompts. The second row illustrates grasp predictions as rectangles, with centers indicating the gripper's grasp point. Red edges represent the predicted gripper opening, while blue edges denote the fixed gripper width. The third row displays grasp position heatmaps from RoG-SAM, highlighting the likelihood of each pixel being a grasp center. The fourth row provides ground truth heatmaps.

The results show that RoG-SAM's predicted grasp positions align closely with actual grasp locations. Additionally, the Cornell dataset annotates grasp affordances using discontinuous line segments, as seen in the bowl and glasses annotations in the fourth row. Despite this, RoG-SAM produces smooth and coherent heatmap predictions, maintaining accuracy across these gaps.

We conducted additional experiments on the large-scale Jacquard grasp dataset to further assess RoG-SAM's generalization capabilities. The evaluation was performed across various Jaccard index thresholds. As shown in Table II, RoG-SAM achieved a grasp accuracy of 94.6% at a Jaccard threshold of 0.25, outperforming other RGB and instance-level methods by approximately 2% and achieving performance comparable to

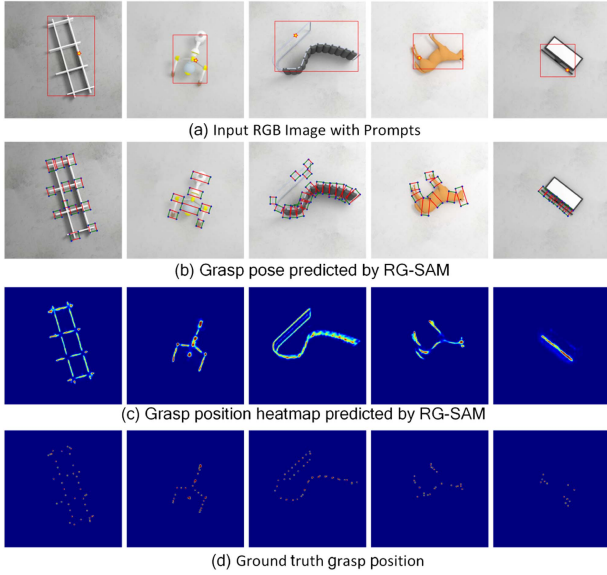


Fig. 6. Detection results on Jacquard dataset.

depth-based methods. Additionally, as the Jaccard threshold increased, the performance of RoG-SAM exhibited a more gradual decline compared to other methods. For instance, at a Jaccard threshold of 0.25, RoG-SAM's accuracy was 1.7% higher than Grasp-Det [6], and this margin expanded to 4.6% at a threshold of 0.35.

Fig. 6 presents representative predictions from Jacquard dataset. Objects in the Jacquard dataset often have intricate structures, making grasp prediction more challenging. For example, in the first column, a grid-like object offers multiple feasible grasp poses along the edges of each grid cell, all of which RoG-SAM accurately predicted with comprehensive coverage. The second column features a tridentate object, requiring prediction of different grasp widths for different sections to secure a grip, which RoG-SAM effectively achieved. Objects in the third and fourth columns also demonstrated a wide range of grasp poses, and RoG-SAM produced precise predictions, with heatmaps closely aligning with the ground truth grasp quality.

Besides, the Jacquard dataset's annotations do not always represent optimal grasp poses. For instance, in the fifth column, the monitor's annotated grasp suggests gripping along the screen edges, which is impractical. In contrast, RoG-SAM proposed grasping the monitor stand, offering a more effective solution. This highlights RoG-SAM's ability to generate consistent and practical grasp patterns from extensive training data, even when annotations are limited or suboptimal.

To assess the impact of different configurations within RoG-SAM, we performed ablation studies on Jacquard dataset, exploring various combinations of input prompts and training strategies, see Table III. The results show that the highest accuracy was achieved by utilizing both box and point prompts together, compared to using either prompt type alone. Notably, using only box prompt resulted in a 0.4% accuracy improvement over using only point prompt, as shown in rows 1 and 2. This improvement likely due to the additional target information provided by bounding boxes, as discussed in [16].

TABLE III
ABLATION STUDY OF BOX/POINT PROMPTS AND FINE-TUNING STRATEGIES (FULL VS. ADAPTER) ON JACQUARD DATASET

Prompt	Training strategy	Accuracy (%)	No. params. (M)
Box	Adapter fine-tuning	94.0	31.3
Point	Adapter fine-tuning	93.6	31.3
Box + Point	SAM Pre-trained	33.8	—
Box + Point	Full Parameter	95.0	110.3
Box + Point	Adapter Fine-tuning	94.6	31.3

We further evaluated RoG-SAM's performance under three different conditions: using SAM's pre-trained weights without training, fine-tuning with adapters, and full parameter training, as detailed in rows 3, 4, and 5. When using SAM's pre-trained weights without transfer learning, RoG-SAM reached an accuracy of only 33.8% on Jacquard dataset, which has a substantial drop compared with fine-tuning or full parameter training. These results suggest that SAM's zero-shot capability is less effective for specialized downstream tasks like grasp detection than it is for natural image segmentation, as supported by recent studies [17]. Transfer learning proves to be necessary for adapting SAM to perform effectively in grasp detection scenarios.

Training RoG-SAM with all parameters achieves a 95.0% grasp success rate, showing a 0.4% gain over fine-tuning with adapters. However, this approach increased the number of trainable parameters by 71%, bringing the total to 110.3 million. Fine-tuning with adapters, therefore, offers a nearly equivalent performance while updating fewer than one-third of the parameters, keeping the training burden comparable to convolution-based detection networks. Overall, RoG-SAM surpasses existing optimized grasping methods and demonstrates greater training efficiency compared to the original SAM.

We also evaluated RoG-SAM in cluttered environments using two instance-level datasets: GraspNet-1Billion and OCID. In the GraspNet-1Billion dataset, objects are concentrated in the center of the scene, we employed the GT Prompt as box prompts for visualization. In contrast, objects in OCID dataset are more dispersed throughout the scene with more pronounced inter-class differences. Thus, we used the Detected Prompt for visualization, as illustrated in Fig. 8. On GraspNet-1Billion, we utilized the raw RGB images and annotations at their original resolution (1280×720) for both training and evaluation, without applying any preprocessing steps that might alter the grasp labels—such as square cropping [47] or label filtering [48]. We made predictions for each instance in the test set containing valid grasp poses (i.e., where the grasp pose centers lie within the object's semantic mask) and reported the accuracy across seen, similar, and novel splits, as presented in Table IV. Underlined values indicate the second-highest performance. Fig. 7 further illustrates sample predictions from each split: columns 1 and 2 correspond to scenes from the seen split, columns 3 and 4 to the similar split, and columns 5 and 6 to the novel split. Notably, RoG-SAM demonstrates strong generalization capabilities on GraspNet-1Billion, as evidenced by its competitive accuracy and smaller performance gap across test set splits. Moreover, RoG-SAM provides reasonable grasp predictions for objects with few or no grasp annotations, such as the red cup in the first column

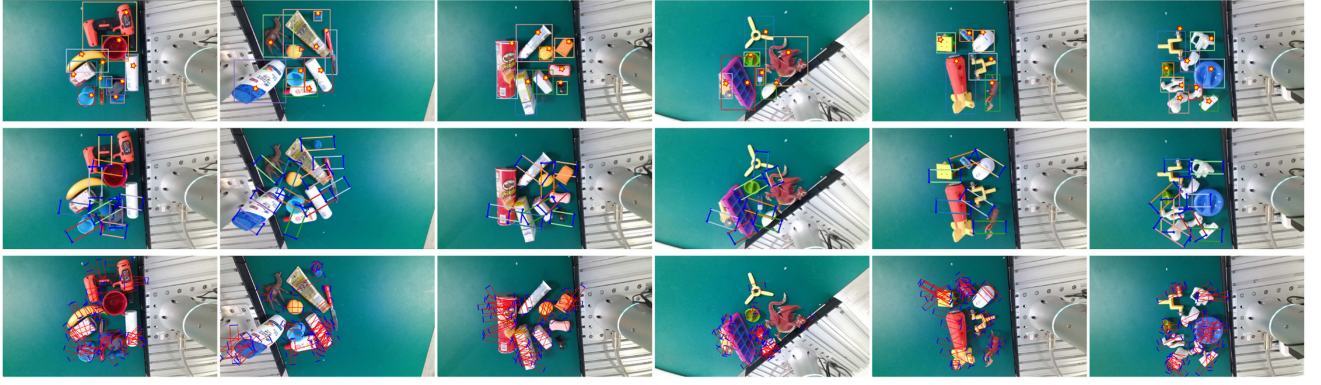


Fig. 7. RoG-SAM's prediction results on GraspNet-1Billion dataset for three splits: Seen (columns 1-2), Similar (columns 3-4), and Novel (columns 5-6). The first row displays RoG-SAM's box prompts and point prompts, with each instance's box prompt shown in a unique color. In the second row, RoG-SAM's predicted grasp poses are presented; to maintain clarity, we visualize up to the top two grasp poses with the highest confidence for each instance. The grasp pose boxes are color-coded to match their corresponding prompts. The third row shows grasp pose labels represented as rectangles, which are generated and visualized using GraspNet-1Billion API.

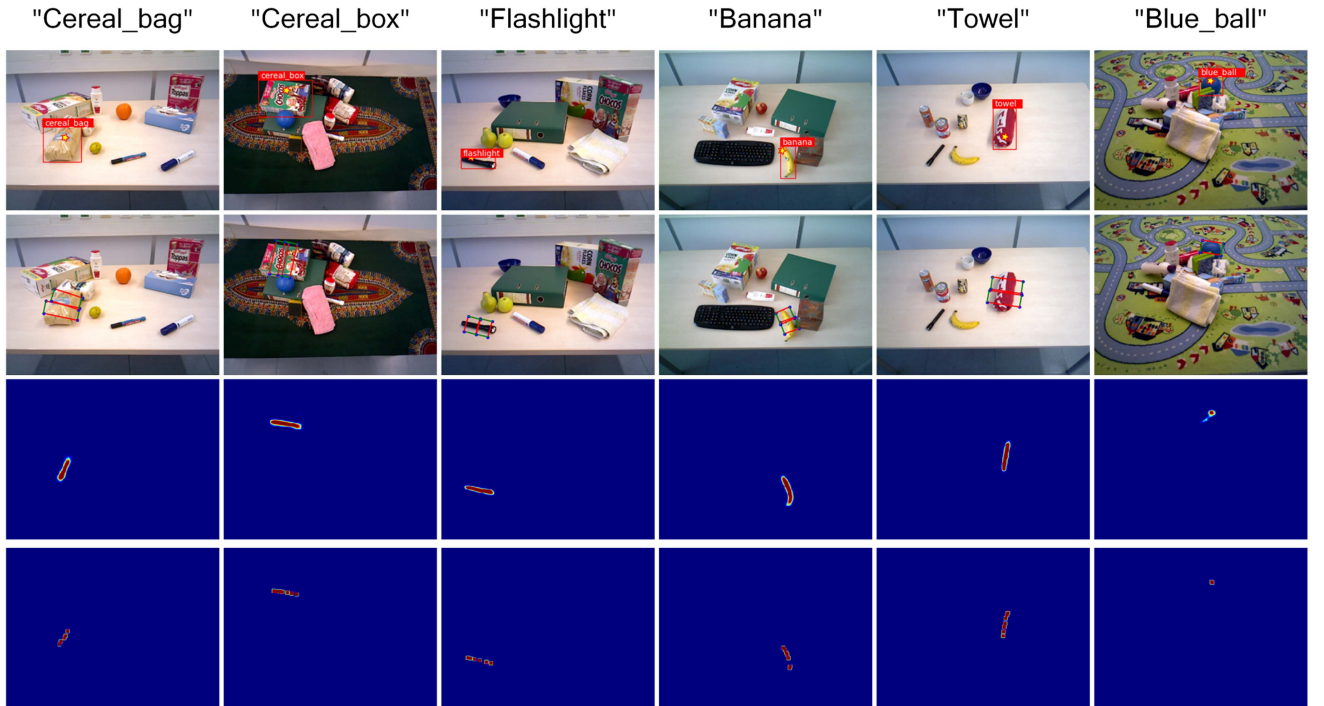


Fig. 8. Detection results on OCID dataset. The input text prompts are open-set, allowing for descriptions based on object appearance, category, or function. The first row shows our input prompts, the second row displays grasp poses predicted by RoG-SAM, and the third and fourth rows present the predicted and ground truth affordance heatmaps, respectively.

TABLE IV
COMPARISON OF RoG-SAM WITH OTHER RECTANGLE-BASED GRASP
DETECTION METHODS ON GRASPNET-1BILLION

Method	Input	Accuracy(%)				IA
		Seen	Similar	Novel	Overall	
Multi-grasp [50]	RGBD	82.7	77.8	72.7	77.7	✗
GG-CNN2 [8]	D	83.0	79.4	76.3	79.6	✗
GR-ConvNet [2]	RGBD	—	—	—	85.6	✗
Grasp RPN [51]	RGD	—	—	—	86.3	✗
CGNet [52]	RGB	—	—	—	87.9	✗
MSGNet [47]	RGBD	—	—	—	89.7	✓
GR-ConvNet2 [48]	RGBD	97.9	96.0	90.5	94.8	✗
RoG-SAM	RGB	<u>93.6</u>	<u>90.9</u>	<u>89.1</u>	<u>91.2</u>	✓

and the black model at the bottom-right of the third column. We further evaluated RoG-SAM on GraspNet-1Billion using different quality scores for a more comprehensive analysis, as shown in Table V. The parameter μ represents the quality threshold of a grasp, with lower μ values indicating higher-quality grasps threshold. We reported AP values for $\mu = 0.8$, $\mu = 0.4$, and the average AP_μ across $\mu = 0.2$ to $\mu = 1.0$. For consistency, we conducted these tests using the rectangle grasp format with the Jaccard metric, unlike some works that adopt a 6-DoF grasp evaluation [49]. The results reflect RoG-SAM's performance using images captured from both Realsense/Kinect cameras (indicated as Realsense/Kinect in Table V).

TABLE V
ROG-SAM PERFORMANCE ACROSS SPLITS AND QUALITY SCORES ON GRASPNET-1BILLION

Split	AP	AP _{0.8}	AP _{0.4}
Seen	82.8/84.3	86.0/89.2	74.5/76.9
Similar	80.5/82.6	83.9/87.7	68.3/70.7
Novel	73.0/74.6	81.1/84.5	65.2/66.8

TABLE VI
IMPACT OF ROG-SAM'S BACKBONE ON INFERENCE TIME AND ACCURACY ON GRASPNET-1BILLION

Backbone	Params. (M)	Inference Time (ms)		Accuracy (%)
		Grounding	Prediction	
ViT-B	110	162	176	91.2
ViT-L	328	162	370	93.5
ViT-H	658	162	472	94.4

TABLE VII
PERFORMANCE COMPARISON WITH RELATED WORKS ON OCID DATASET

Method	Accuracy(%)	Year	Instance-aware method
OCID [6]	89.0	2022	Segmentation
Efficient-RGB [53]	74.7	2023	✗
Efficient-RGBD [53]	76.4	2023	✗
SSG [45]	92.0	2023	2D detector
Grasp-anything [1]	74.0	2024	✗
RoG-SAM	<u>90.1</u>	—	Prompt

Building on these evaluations, we conducted an ablation study to investigate the impact of different RoG-SAM image encoders on performance and inference time, testing three backbones: ViT-B, ViT-L, and ViT-H. The results are summarized in Table VI. Inference time is divided into two stages: **Grounding**, which measures the time required to convert language prompts into position prompts, and **Prediction**, which accounts for the time taken to generate grasp pose predictions. As the results indicate, the largest backbone, ViT-H, achieves a 3.2% accuracy improvement over the default ViT-B, but at a cost of a $2.7\times$ increase in inference time and a $6\times$ increase in trainable parameters. In contrast, the default ViT-B provides a good trade-off between performance and computational efficiency, making it a practical choice for real-time applications. Considering real-time performance and training costs, we recommend using ViT-B or ViT-L for practical tasks.

We also evaluated RoG-SAM on OCID grasp dataset, which includes a wider variety of object categories. As detailed in Table VII, our approach achieved a grasp detection accuracy of 90.1%, surpassing convolution-based global detection methods [1], [53] by 15.4% and 16.1%, respectively, and exceeding the global detection method with semantic segmentation post-processing [6] by 1.1%. However, it was 1.9% lower than the multi-stage method SSG [45], which employs a complex pipeline involving object detection, coefficient prediction, mask combination, and post-filtering. In contrast, RoG-SAM directly predicts grasps from RGB images using multi-head segmentation. Moreover, RoG-SAM is a language-driven grasp detection approach capable of open-set instance-level detection, unlike

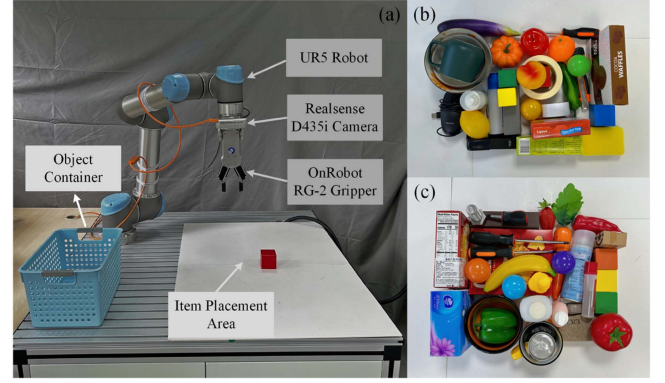


Fig. 9. (a) The UR5 robot grasping experimental setup. For each experiment, the robot iteratively plans grasp poses from RGB images and attempts to lift and transport the objects to the Object Container. (b) A set of 30 objects used for training, referred to as “seen” objects. (c) A set of 30 objects used for testing, referred to as “unseen” objects.

methods that rely on global predictions [1], [53] or visual detectors limited to predefined categories [6], [45].

Fig. 8 showcases several detection results on OCID dataset. During testing, we utilized the detected bounding box and point prompts for RoG-SAM. The input text prompts were open-set, allowing for descriptions based on instance color, category, or function. Various text prompts could be employed to focus on a specific instance, such as using “green_box” instead of “cereal_box” in the second row. With prompts generated by the grounding detector, RoG-SAM delivers accurate grasp suggestions even in challenging scenarios, such as samples with textured backgrounds, as seen in column 2, and targets partially occluded by other objects, as illustrated in column 6.

V. EXPERIMENTS IN REAL-WORLD ENVIRONMENTS

This section details the experimental setup and the results of grasping tests conducted with the UR5 robot in an indoor laboratory environment.

A. Experimental Setup

In addition to evaluating RoG-SAM on public grasping datasets, we conducted real-world grasping experiments using a UR5 robot. The robot was equipped with an OnRobot RoG-2 parallel gripper for manipulation tasks and a Realsense D435i Depth camera for visual perception. Our model was deployed on an Ubuntu 20.04 system, running on an a GeForce RTX 2060 GPU. The experimental setup is illustrated in Fig. 9. The UR5 robot arm begins in an initial position, vertically aligned 600 mm above the placement zone. An object container is positioned at the side of the workbench for storing the grasped items.

To validate the practical utility of RoG-SAM, we selected 60 common grasping targets from Cornell and OCID datasets. These objects were chosen to align with the requirements of potential application scenarios, including items for service robots (e.g., fruits, tableware), logistics warehouses (e.g., packaging boxes, electronic devices), and manufacturing tools (e.g., screwdrivers, wrenches). Of these, 30 objects were used as training

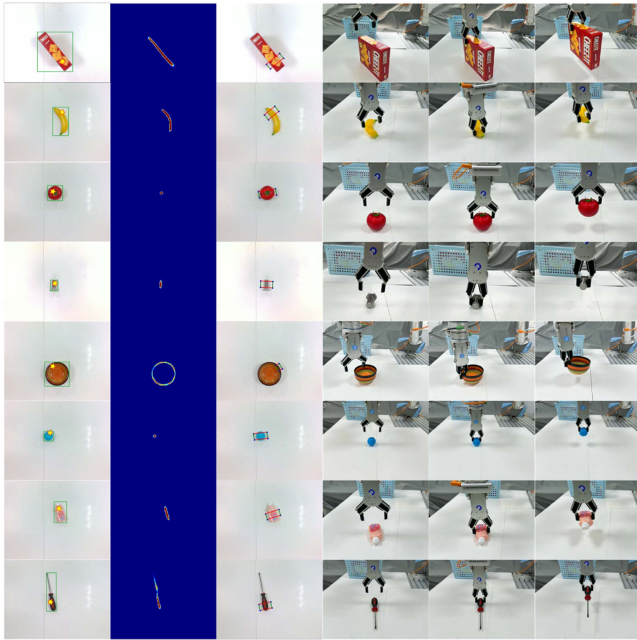


Fig. 10. Representative Detection Results and Screenshots of the UR5 Grasping Process for Single-Object Experiments. Each row corresponds to a different object. Columns 1-3 visualize the input box and point prompts, grasp position heatmaps, and predicted grasp poses, while Columns 4-6 illustrate the grasping process. The input text prompts, from top to bottom, are: “cracker box,” “banana,” “tomato,” “bolt,” “bowl,” “blue ball,” “plastic container,” and “screwdriver”.

samples, as illustrated in Fig. 9(b), while the remaining 30 unseen objects were reserved for testing the model’s performance, as shown in Fig. 9(c). The training dataset for our grasping model consisted of 1,000 samples randomly sampled from Cornell and OCID datasets, along with 100 manually annotated images from our lab environment containing objects depicted in Fig. 9(b).

In our grasping experiments, the workflow for each attempt was as follows: an RGB image was captured by the camera, a text prompt was provided by a human operator, RoG-SAM generated a grasp pose prediction, and the UR5 robot executed the grasp based on the predicted end-effector configuration. A grasp was considered successful if target object was successfully placed into the container. Building on established methodologies [2], [54], [55], [56], our experimental design incorporated two primary task scenarios—Single and Clutter—aimed at evaluating RoG-SAM’s overall grasping performance in comparison to existing approaches.

B. Single-Object Grasping Experiments

In the single-object experiments, each object from Fig. 9(c) was randomly placed in the designated area, with 12 tests conducted per object, varying both position and orientation. The object box and point prompts were generated using grounding detector, as indicated by green bounding boxes and yellow stars in the first column of Fig. 10. RoG-SAM then predicted the grasp position heatmap, width heatmap, and angle heatmap from the captured images. For clarity, the position heatmap is displayed in the second column. The final grasp pose was determined by

TABLE VIII
COMPARISON OF REAL-WORLD ROBOTIC ARM GRASPING PERFORMANCE

Method	Platform	Single (%)	Clutter (%)
Dex-Net2 [55]	ABB YuMi	80.0	—
GG-CNN2 [8]	Kinova Mico	92.0	87.0
GR-ConvNet [2]	Baxter Robot	95.4	93.5
DSQNet [57]	Franka Emika	93.0	—
Anygrasp [56]	Universal Robots	—	93.3
Gao et al. [54]	Kinova Gen2	96.7	94.1
RoG-SAM	Universal Robots	97.5	94.6



Fig. 11. Detection results of RoG-SAM in cluttered scenes combining outcomes for each object.

selecting the prediction with the highest position confidence, as shown in the third column of Fig. 10. The fourth through sixth columns provide snapshots of grasping process from a fixed camera perspective, capturing the gripper 50 mm above the grasp pose, at the moment of gripper closure, and after lifting the object 70 mm post-grasp.

From these examples, we observed that although affordance map suggested multiple potential grasp points, RoG-SAM tend to select grasp locations on parts with a more uniform width distribution—such as the tail of banana, the shaft of bolt, and the central of plastic container—rather than the base of bolt or the protruding sections of plastic container. This behavior aligns closely with the preferences observed in human operators.

To further evaluate RoG-SAM’s real-world grasping performance, we compared it against other grasping algorithms on previously unseen objects, as detailed in Table VIII. Out of 360 total grasp attempts, our method successfully completed 351 grasps, resulting in an accuracy rate of 97.5%. This outcome demonstrates RoG-SAM’s capability to effectively handle novel objects and highlights its superior performance relative to other methods in comparable grasping tasks.

C. Cluttered-Object Grasping Experiments

In settings such as industrial manufacturing, household services, and warehouse sorting, objects are often arranged in variable, disordered, and unstructured ways [58], [59]. To better replicate these conditions, we conducted cluttered-object grasping experiments. For these experiments, we randomly selected 5-10 objects from those depicted in Fig. 9(c) and placed them within the workspace. We provided the robot with language prompts for each object, guiding it to sequentially grasp objects until all prompts were performed.

Fig. 11 illustrate two typical cluttered scenarios, featuring densely packed items such as screwdriver, cracker box, and



Fig. 12. Examples of failed grasps during RoG-SAM's execution on UR5 robot.

fruits. The second column presents box prompts and point prompts generated by the grounding detector for each object. The affordance heatmap predicted by RoG-SAM is overlaid on RGB image in the third column, while up to four feasible grasp poses for each object are displayed in the fourth column. The results demonstrate that RoG-SAM effectively focuses on the objects specified by user prompt, even in scenarios where multiple objects are stacked. For example, in the second row, where a blue ball partially obscures a cracker box, RoG-SAM accurately predicts the affordance for both blue ball and the visible part of cracker box respectively, offering high-quality grasp suggestions for each object.

In the cluttered-object experiments, we conducted 35 trials involving a total of 276 objects. The final grasping results are presented in the fourth column of Table. VIII. RoG-SAM achieved an average grasp success rate of 94.6% in cluttered scenes, demonstrates its strong ability to handle complex scenarios involving stacked objects, outperforming existing methods under same experimental conditions.

D. Failure Case Analysis

The primary causes of grasping failures can be summarized as follows:

- 1) *Gripper obstruction by surrounding objects*: Some grasping attempts failed because the gripper was obstructed by nearby objects during its approach. For instance, in the first row of Fig. 12, the robotic gripper was blocked by an adjacent glasses case while attempting to grasp the screwdriver.
- 2) *Inaccurate depth estimation*: In some cases, inaccuracies in depth estimation, caused by minor misalignments between color and depth frames, led to incorrect grasping heights. For example, in the second row of Fig. 12, the depth value of bottom of the cup was used for grasping, resulting in a collision with the object.

VI. CONCLUSION

This paper presents RoG-SAM, a language-driven, instance-level grasp detection model based on the SAM framework. We incorporated a grounding detector, which generates sparse 2D prompts to guide grasp pose estimation process for specific instances. To improve training efficiency, we implemented adapter modules, reducing the number of trainable parameters to 28.3% of the full SAM model. Experimental results indicate

that RoG-SAM delivers competitive grasp accuracy across both single-object and cluttered datasets. Moreover, real-world experiments demonstrate our model's generalization capabilities, achieving superior performance on unseen objects in single and cluttered environments. We believe RoG-SAM holds potential for applications in industrial automation, household robotics, and warehouse logistics.

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